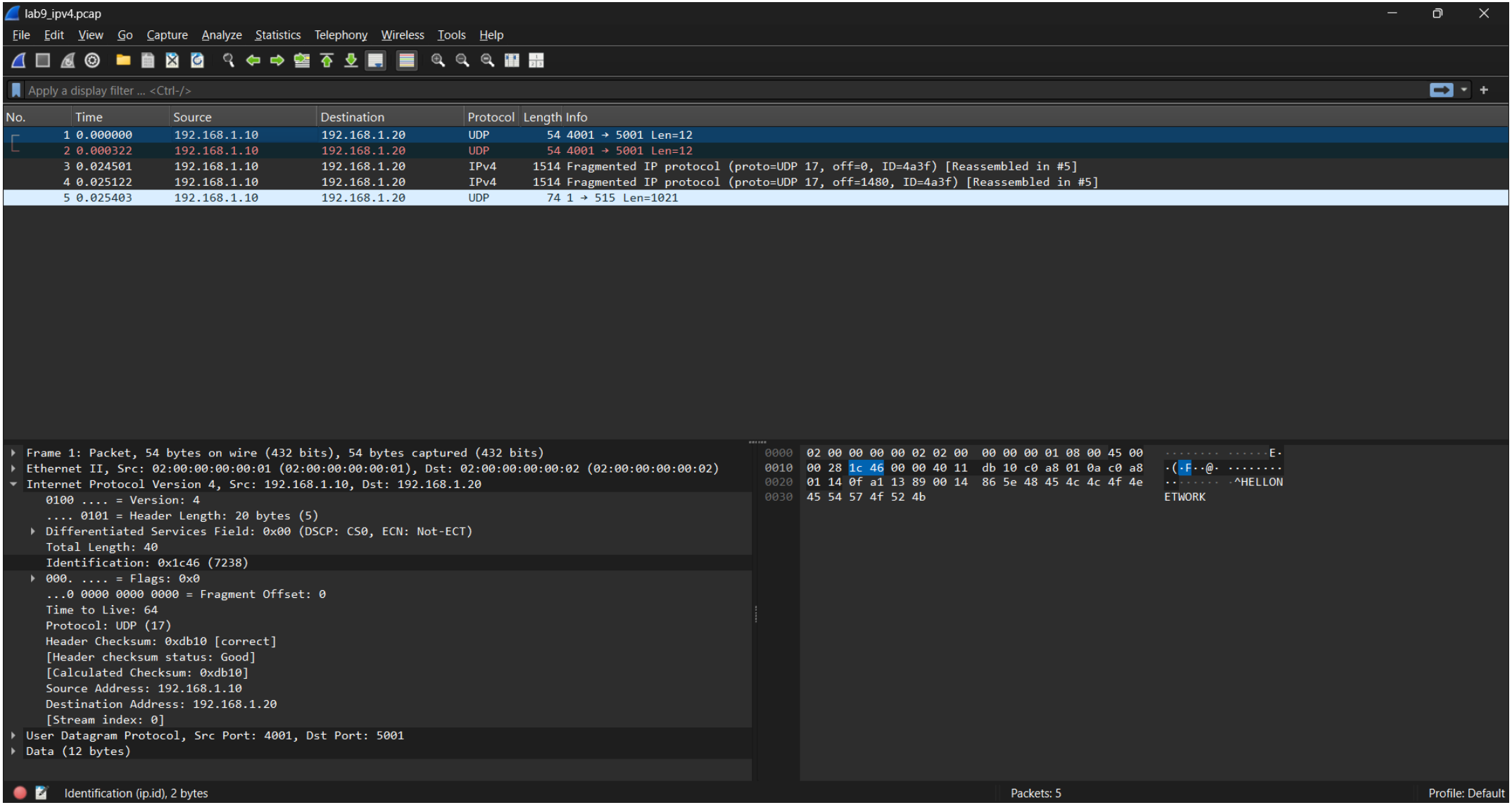
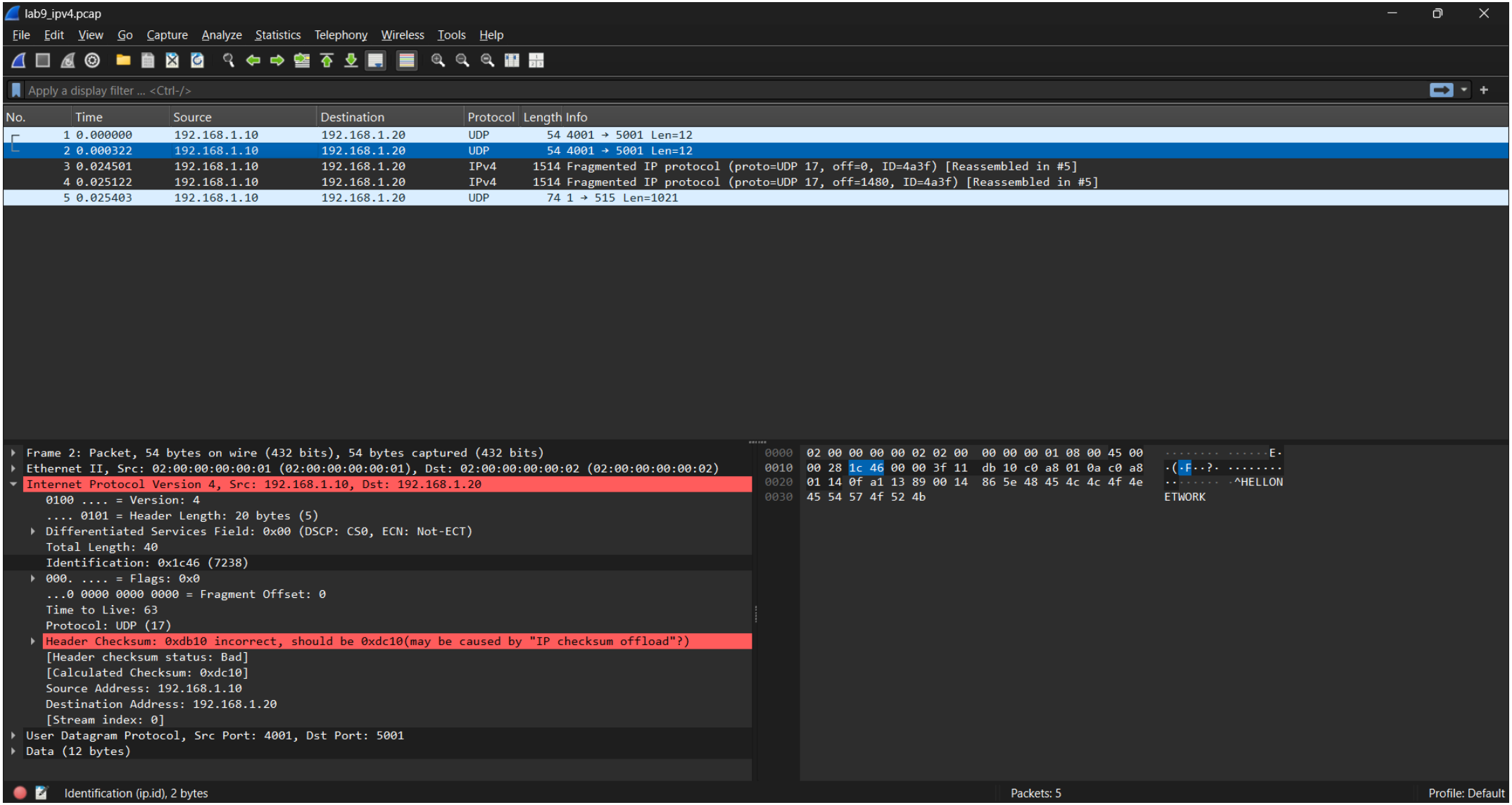
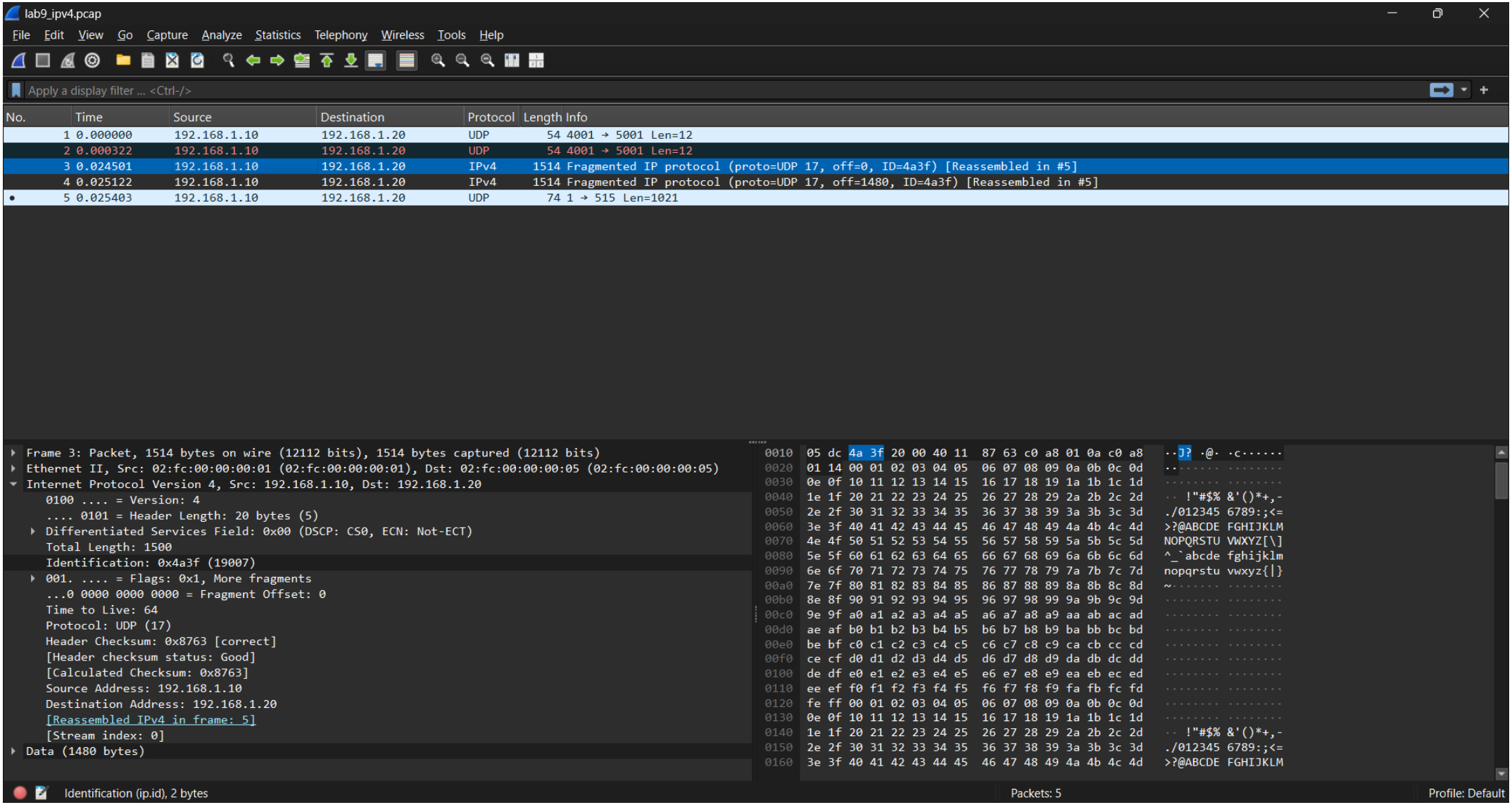
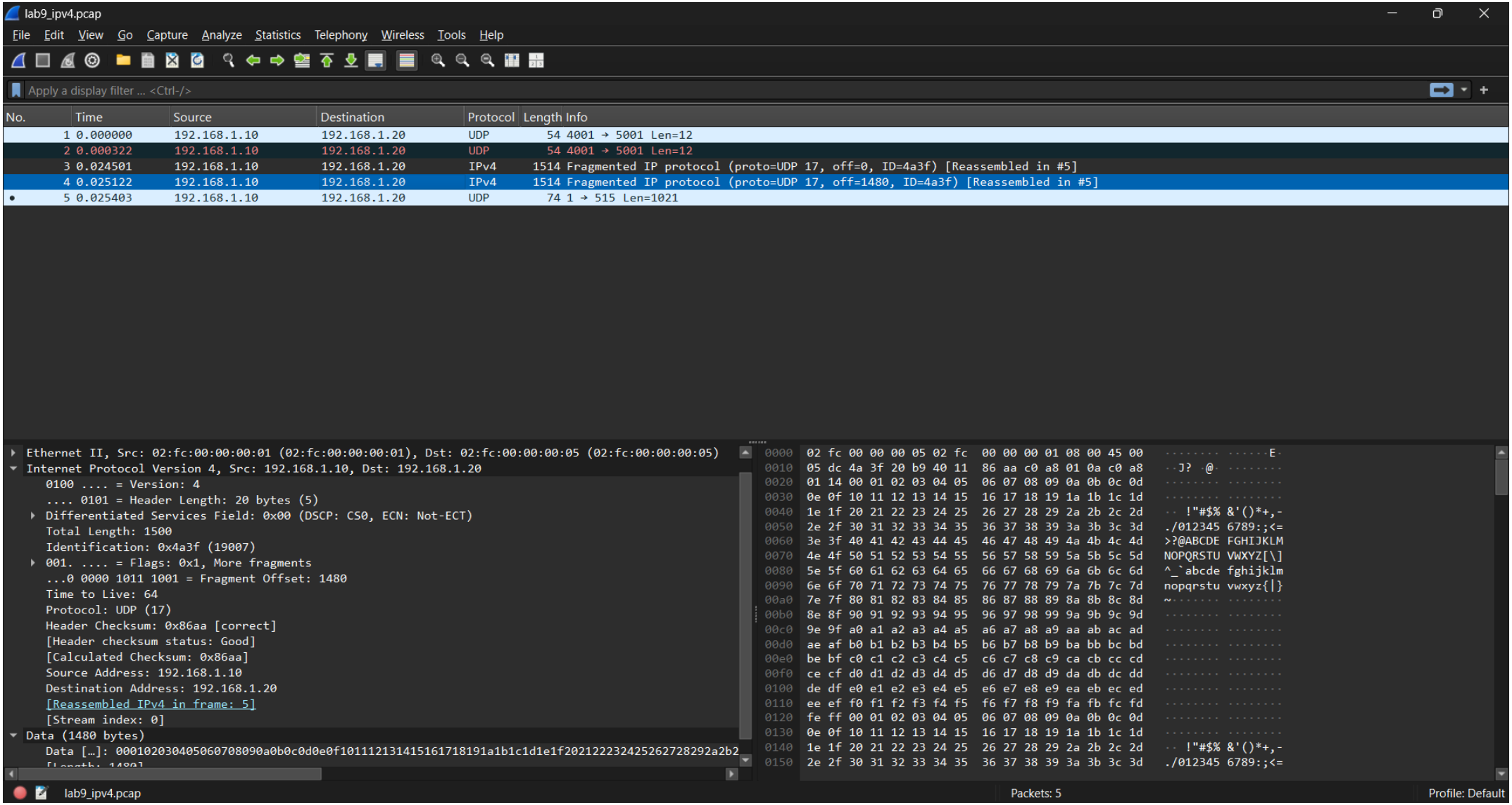
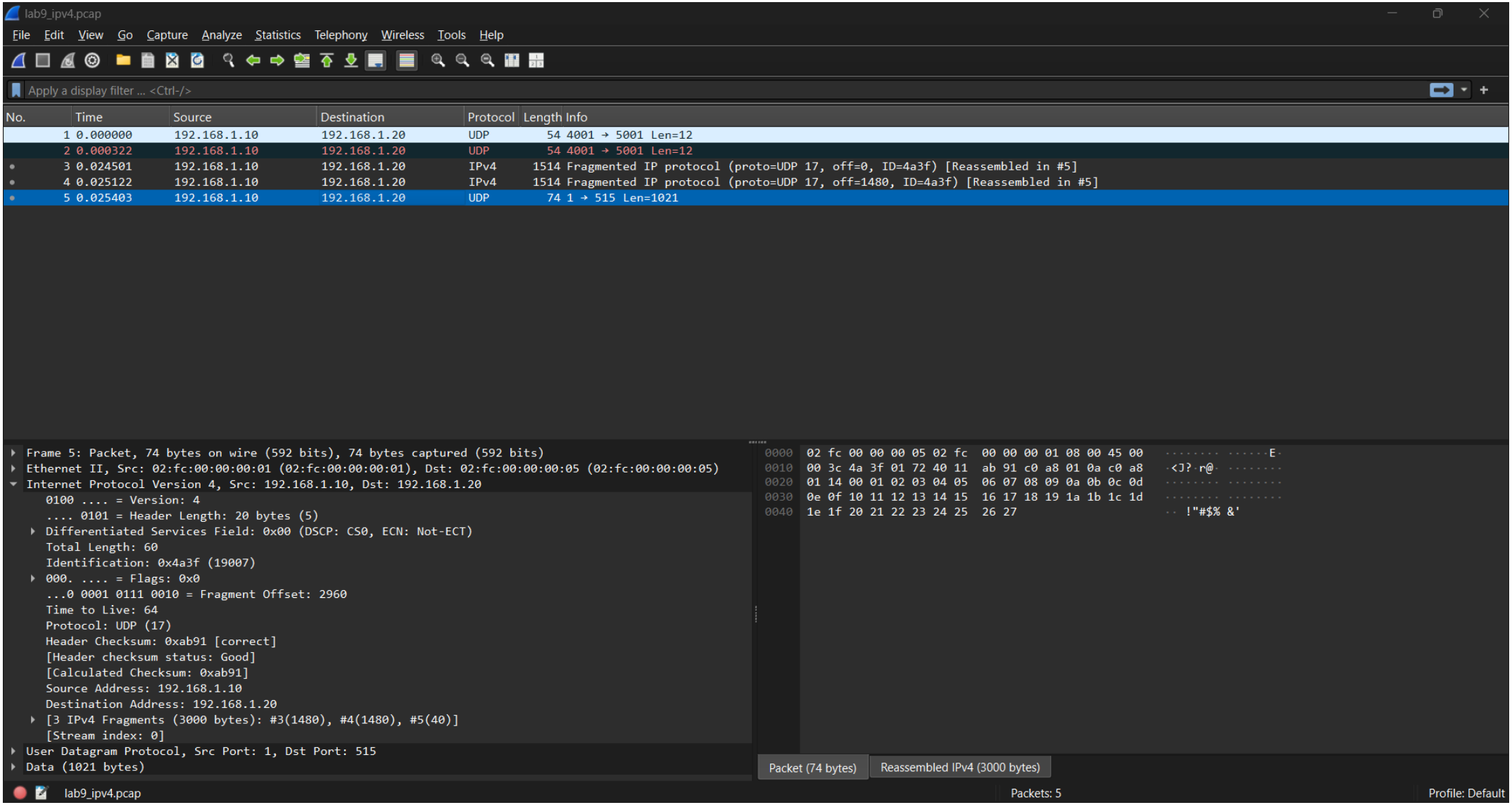
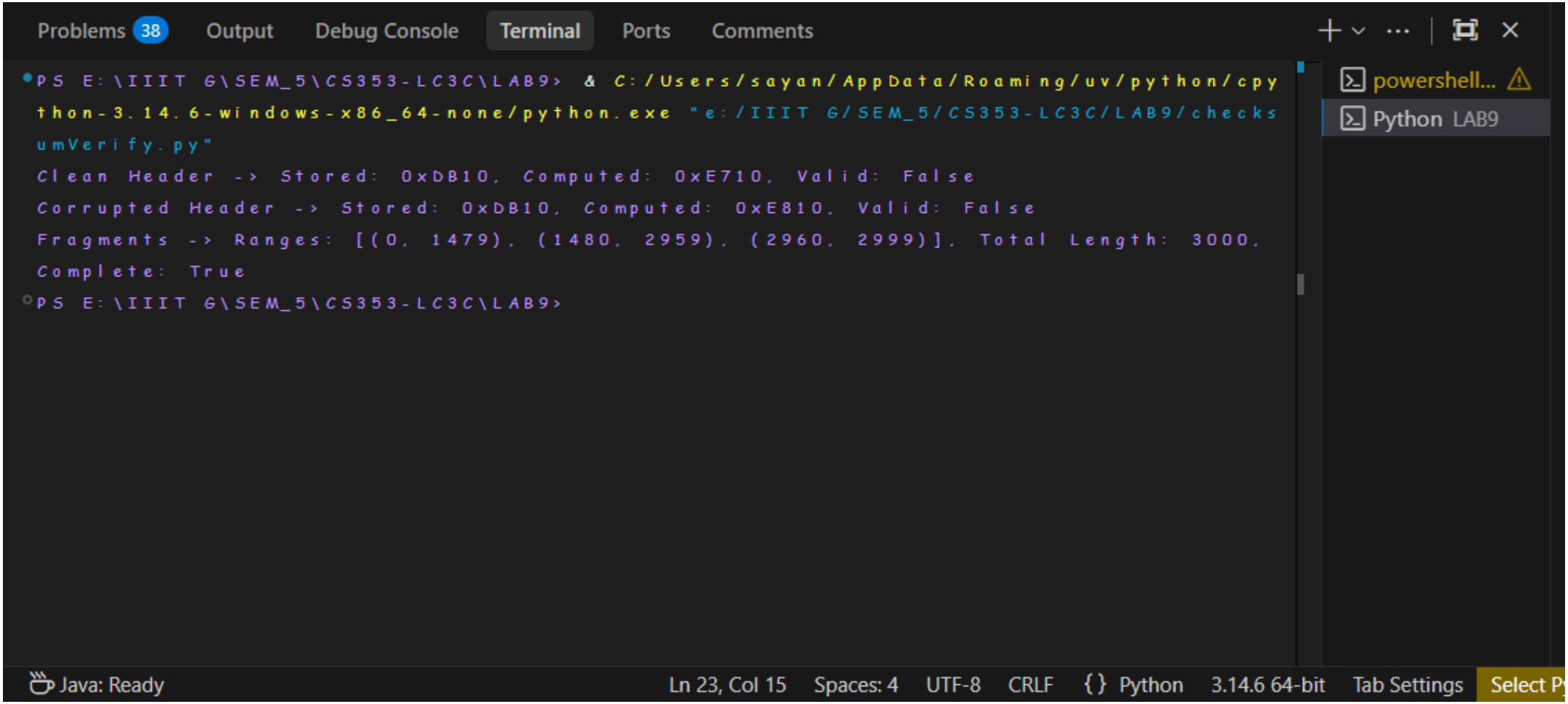

lab9_ipv4.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
2 0.000322 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
3 0.024501 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=0, ID=4a3f) [Reassembled in #5]
4 0.025122 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=1480, ID=4a3f) [Reassembled in #5]
5 0.025403 192.168.1.10 192.168.1.20 UDP 74 1 → 515 Len=1021
Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
Ethernet II, Src: 02:00:00:00:00:01 (02:00:00:00:00:01), Dst: 02:00:00:00:00:02 (02:00:00:00:00:02)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.20
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 40
Identification: 0x1c46 (7238)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0xdb10 [correct]
[Header checksum status: Good]
[Calculated Checksum: 0xdb10]
Source Address: 192.168.1.10
Destination Address: 192.168.1.20
[Stream index: 0]
User Datagram Protocol, Src Port: 4001, Dst Port: 5001
Data (12 bytes)
0000 02 00 00 00 00 02 02 00 00 00 00 01 08 00 45 00
0010 00 28 1c 46 00 00 40 11 db 10 c0 a8 01 0a c0 a8
0020 01 14 0f a1 13 89 00 14 86 5e 48 45 4c 4c 4f 4e
0030 45 54 57 4f 52 4b
^HELLON
ETWORK
Identification (ip.id), 2 bytes
Packets: 5
Profile: Default


lab9_ipv4.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
2 0.000322 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
3 0.024501 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=0, ID=4a3f) [Reassembled in #5]
4 0.025122 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=1480, ID=4a3f) [Reassembled in #5]
5 0.025403 192.168.1.10 192.168.1.20 UDP 74 1 → 515 Len=1021
Frame 2: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
Ethernet II, Src: 02:00:00:00:00:01 (02:00:00:00:00:01), Dst: 02:00:00:00:00:02 (02:00:00:00:00:02)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.20
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 40
Identification: 0x1c46 (7238)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 63
Protocol: UDP (17)
Header Checksum: 0xdb10 incorrect, should be 0xdc10(may be caused by "IP checksum offload"?)
[Header checksum status: Bad]
[Calculated Checksum: 0xdc10]
Source Address: 192.168.1.10
Destination Address: 192.168.1.20
[Stream index: 0]
User Datagram Protocol, Src Port: 4001, Dst Port: 5001
Data (12 bytes)
0000 02 00 00 00 00 02 02 00 00 00 00 01 08 00 45 00
0010 00 28 1c 46 00 00 3f 11 db 10 c0 a8 01 0a c0 a8
0020 01 14 0f a1 13 89 00 14 86 5e 48 45 4c 4c 4f 4e
0030 45 54 57 4f 52 4b
^HELLON
ETWORK
Identification (ip.id), 2 bytes
Packets: 5
Profile: Default


lab9_ipv4.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
2 0.000322 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
3 0.024501 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=0, ID=4a3f) [Reassembled in #5]
4 0.025122 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=1480, ID=4a3f) [Reassembled in #5]
5 0.025403 192.168.1.10 192.168.1.20 UDP 74 1 → 515 Len=1021
Frame 3: Packet, 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits)
Ethernet II, Src: 02:fc:00:00:00:01 (02:fc:00:00:00:01), Dst: 02:fc:00:00:00:05 (02:fc:00:00:00:05)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.20
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1500
Identification: 0x4a3f (19007)
001. .... = Flags: 0x1, More fragments
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0x8763 [correct]
[Header checksum status: Good]
[Calculated Checksum: 0x8763]
Source Address: 192.168.1.10
Destination Address: 192.168.1.20
[Reassembled IPv4 in frame: 5]
[Stream index: 0]
Data (1480 bytes)
0010 05 dc 4a 3f 20 00 40 11 87 63 c0 a8 01 0a c0 a8
0020 01 14 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d
0060 3e 3f 40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d
0070 4e 4f 50 51 52 53 54 55 56 57 58 59 5a 5b 5c 5d
0080 5e 5f 60 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d
0090 6e 6f 70 71 72 73 74 75 76 77 78 79 7a 7b 7c 7d
00a0 7e 7f 80 81 82 83 84 85 86 87 88 89 8a 8b 8c 8d
00b0 8e 8f 90 91 92 93 94 95 96 97 98 99 9a 9b 9c 9d
00c0 9e 9f a0 a1 a2 a3 a4 a5 a6 a7 a8 a9 aa ab ac ad
00d0 ae af b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 ba bb bc bd
00e0 be bf c0 c1 c2 c3 c4 c5 c6 c7 c8 c9 ca cb cc cd
00f0 ce cf d0 d1 d2 d3 d4 d5 d6 d7 d8 d9 da db dc dd
0100 de df e0 e1 e2 e3 e4 e5 e6 e7 e8 e9 ea eb ec ed
0110 ee ef f0 f1 f2 f3 f4 f5 f6 f7 f8 f9 fa fb fc fd
0120 fe ff 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d
0130 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0140 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d
0150 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d
0160 3e 3f 40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d
Identification (ip.id), 2 bytes
Packets: 5
Profile: Default


lab9_ipv4.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
2 0.000322 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
3 0.024501 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=0, ID=4a3f) [Reassembled in #5]
4 0.025122 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=1480, ID=4a3f) [Reassembled in #5]
5 0.025403 192.168.1.10 192.168.1.20 UDP 74 1 → 515 Len=1021
Ethernet II, Src: 02:fc:00:00:00:01 (02:fc:00:00:00:01), Dst: 02:fc:00:00:00:05 (02:fc:00:00:00:05)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.20
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1500
Identification: 0x4a3f (19007)
001. .... = Flags: 0x1, More fragments
...0 0000 1011 1001 = Fragment Offset: 1480
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0x86aa [correct]
[Header checksum status: Good]
[Calculated Checksum: 0x86aa]
Source Address: 192.168.1.10
Destination Address: 192.168.1.20
[Reassembled IPv4 in frame: 5]
[Stream index: 0]
Data (1480 bytes)
Data […]: 000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2
0000 02 fc 00 00 00 05 02 fc 00 00 00 01 08 00 45 00
0010 05 dc 4a 3f 20 b9 40 11 86 aa c0 a8 01 0a c0 a8
0020 01 14 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d
0060 3e 3f 40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d
0070 4e 4f 50 51 52 53 54 55 56 57 58 59 5a 5b 5c 5d
0080 5e 5f 60 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d
0090 6e 6f 70 71 72 73 74 75 76 77 78 79 7a 7b 7c 7d
00a0 7e 7f 80 81 82 83 84 85 86 87 88 89 8a 8b 8c 8d
00b0 8e 8f 90 91 92 93 94 95 96 97 98 99 9a 9b 9c 9d
00c0 9e 9f a0 a1 a2 a3 a4 a5 a6 a7 a8 a9 aa ab ac ad
00d0 ae af b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 ba bb bc bd
00e0 be bf c0 c1 c2 c3 c4 c5 c6 c7 c8 c9 ca cb cc cd
00f0 ce cf d0 d1 d2 d3 d4 d5 d6 d7 d8 d9 da db dc dd
0100 de df e0 e1 e2 e3 e4 e5 e6 e7 e8 e9 ea eb ec ed
0110 ee ef f0 f1 f2 f3 f4 f5 f6 f7 f8 f9 fa fb fc fd
0120 fe ff 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d
0130 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0140 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d
0150 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d
lab9_ipv4.pcap
Packets: 5
Profile: Default


lab9_ipv4.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
2 0.000322 192.168.1.10 192.168.1.20 UDP 54 4001 → 5001 Len=12
3 0.024501 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=0, ID=4a3f) [Reassembled in #5]
4 0.025122 192.168.1.10 192.168.1.20 IPv4 1514 Fragmented IP protocol (proto=UDP 17, off=1480, ID=4a3f) [Reassembled in #5]
5 0.025403 192.168.1.10 192.168.1.20 UDP 74 1 → 515 Len=1021
Frame 5: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits)
Ethernet II, Src: 02:fc:00:00:00:01 (02:fc:00:00:00:01), Dst: 02:fc:00:00:00:05 (02:fc:00:00:00:05)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.20
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0x4a3f (19007)
000. .... = Flags: 0x0
...0 0001 0111 0010 = Fragment Offset: 2960
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0xab91 [correct]
[Header checksum status: Good]
[Calculated Checksum: 0xab91]
Source Address: 192.168.1.10
Destination Address: 192.168.1.20
[3 IPv4 Fragments (3000 bytes): #3(1480), #4(1480), #5(40)]
[Stream index: 0]
User Datagram Protocol, Src Port: 1, Dst Port: 515
Data (1021 bytes)
0000 02 fc 00 00 00 05 02 fc 00 00 00 01 08 00 45 00
0010 00 3c 4a 3f 01 72 40 11 ab 91 c0 a8 01 0a c0 a8
0020 01 14 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0040 1e 1f 20 21 22 23 24 25 26 27
Packet (74 bytes)
Reassembled IPv4 (3000 bytes)
lab9_ipv4.pcap
Packets: 5
Profile: Default


Problems 38
Output
Debug Console
Terminal
Ports
Comments
PS E:\IIIT G\SEM_5\CS353-LC3C\LAB9> & C:/Users/sayan/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_64-none/python.exe "e:/IIIT G/SEM_5/CS353-LC3C/LAB9/checksumVerify.py"
Clean Header -> Stored: 0xDB10, Computed: 0xE710, Valid: False
Corrupted Header -> Stored: 0xDB10, Computed: 0xE810, Valid: False
Fragments -> Ranges: [(0, 1479), (1480, 2959), (2960, 2999)], Total Length: 3000, Complete: True
PS E:\IIIT G\SEM_5\CS353-LC3C\LAB9>
powershell...
Python LAB9
Java: Ready
Ln 23, Col 15
Spaces: 4
UTF-8
CRLF
{} Python
3.14.6 64-bit
Tab Settings